Robotics Final Project

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***Abstract*—This paper considers the double inverted pendulum on a cart (DIPC) system, deriving dynamics and a control strategy. The control strategy consists of two phases: swing-up and regulation.**

***Index Terms*—double inverted pendulum, control theory, optimal control**

I. Introduction

The DIP system is an underactuated, highly nonlinear and chaotic system, with four equilibrium points (down-down, down-up, up-down, up-up). The aim of this paper is to start from the stable equilibrium point (down-down), then swing-up and stabilize the pendulum at the three unstable equilibrium points. The approach is comprised of two phases, swing-up and regulation. A switching condition is defined, and once satisfied, we switch from swing-up to regulation. This threshold is defined by the domain of attraction (DoA): the set of initial states from which the regulator can "catch" the pendulums. One approach for the swing-up is to shape the pendulums energy. This approach is inherently chaotic, and does not control the trajectory the system takes. It simply drives the pendulum energy to the level of the desired equilibrium point. The pendulum may swing around in a chaotic manner until it enters the DoA and is stabilized by the regulating controller.

II. Problem Statement

The system is described by the following schematic:

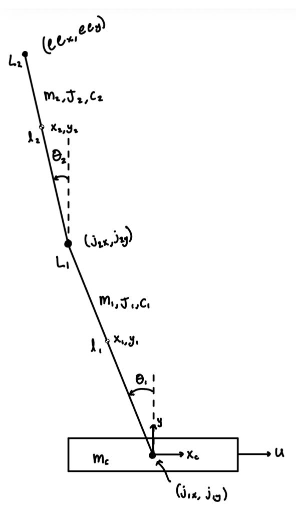

Fig. 1. Double Inverted Pendulum System

and has the following parameters:

TABLE I
System Parameters.

Parameter	Description	Value	Unit
m_c	cart mass	0.4	kg
m_1	link 1 mass	0.15	kg
m_2	link 2 mass	0.15	kg
L_1	link 1 length	0.5	m
L_2	link 2 length	0.5	m
l_1	link 1 l_{com}	$\frac{L_1}{2}$	m
l_2	link 2 l_{com}	$\frac{L_2}{2}$	m
J_1	link 1 MOI	$\frac{m_1 L_1^2}{12}$	kg-m^2
J_2	link 2 MOI	$\frac{m_2 L_2^2}{12}$	kg-m^2
c_c	cart damping coeff.	0	N-s/m
c_1	link 1 damping coeff.	0	N-s/rad
c_2	link 1 damping coeff.	0	N-s/rad
g	gravitational const.	9.81	m/s^2

III. Methods

Describe your methodology, filtering techniques, or experimental setup.

A. System Dynamics

TABLE OF SYSTEM PARAMS The system dynamics were found using Lagrangian mechanics:

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_j} - \frac{\partial L}{\partial q_j} = Q_j \tag{1}$$

where Q_j are the non-conservative forces, and the Lagrangian $L = T - U$. The potential energy U and kinetic energy T are described by the following equations:

$$U = g(m_1 l_1 \cos\theta_1 + m_2(L_1 \cos\theta_1 + l_2 \cos\theta_2)) \tag{2}$$

$$\begin{aligned} T &= T_c + T_1 + T_2 \\ T_c &= \frac{1}{2} m_c \dot{x}_c^2 \\ T_1 &= \frac{1}{2} m_1(\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2} J_1 \dot{\theta}_1^2 \\ T_2 &= \frac{1}{2} m_2(\dot{x}_2^2 + \dot{y}_2^2) + \frac{1}{2} J_2 \dot{\theta}_2^2 \end{aligned} \tag{3}$$

where:

$$
\begin{aligned}
x_1 &= x_c - l_1 \sin\theta_1 \\
y_1 &= l_1 \cos\theta_1 \\
\dot{x}_1 &= \dot{x}_c - l_1 \cos\theta_1 \dot{\theta}_1 \\
\dot{y}_1 &= -l_1 \sin\theta_1 \dot{\theta}_1 \\
x_2 &= x_c - L_1 \sin\theta_1 - l_2 \sin\theta_2 \\
y_2 &= L_1 \cos\theta_1 + l_2 \cos\theta_2 \\
\dot{x}_2 &= \dot{x}_c - L_1 \cos\theta_1 \dot{\theta}_1 - l_2 \cos\theta_2 \dot{\theta}_2 \\
\dot{y}_2 &= -L_1 \sin\theta_1 \dot{\theta}_1 - l_2 \sin\theta_2 \dot{\theta}_2
\end{aligned}
$$

The Euler-Lagrange equations result in:

$$
\begin{aligned}
F - c_c \dot{x}_c = {} & l_2 m_2 \sin(\theta_2)\dot{\theta}_2^2 - \ddot{\theta}_1 (l_1 m_1 \cos(\theta_1) \\
& + L_1 m_2 \cos(\theta_1)) + \dot{\theta}_1 (L_1 m_2 \dot{\theta}_1 \sin(\theta_1) \\
& + l_1 m_1 \dot{\theta}_1 \sin(\theta_1)) + \ddot{x}_c (m_1 + m_2 + m_c) \\
& - l_2 m_2 \ddot{\theta}_2 \cos(\theta_2)
\end{aligned}
\tag{4}
$$

$$
\begin{aligned}
& (L_1 m_2 + l_1 m_1)(g \sin(\theta_1) + \ddot{x}_c \cos(\theta_1)) + c_2 \dot{\theta}_2 \\
& = \ddot{\theta}_1 (J_1 + m_1 l_1^2 + m_2 L_1^2) + \dot{\theta}_1 (c_1 + c_2) \\
& \quad + L_1 l_2 m_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) + L_1 l_2 m_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2)
\end{aligned}
\tag{5}
$$

$$
\begin{aligned}
& m_2 \ddot{\theta}_2 l_2^2 + L_1 m_2 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) l_2 + J_2 \ddot{\theta}_2 + c_2 \dot{\theta}_2 \\
& = L_1 l_2 m_2 \sin(\theta_1 - \theta_2)\dot{\theta}_1^2 + c_2 \dot{\theta}_1 + g l_2 m_2 \sin(\theta_2) \\
& \quad + l_2 m_2 \ddot{x}_c \cos(\theta_2)
\end{aligned}
\tag{6}
$$

These equations are put into compact form:

$$
\mathbf{M}(q)\ddot{\mathbf{q}} + \mathbf{C}(q, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(q) = \mathbf{Q} \tag{7}
$$

and a state space model is defined:

$$
\begin{aligned}
\mathbf{x} &= \begin{bmatrix} x_c & \dot{x}_c & \theta_1 & \dot{\theta}_1 & \theta_2 & \dot{\theta}_2 \end{bmatrix}^T \\
\dot{\mathbf{x}} &= \begin{bmatrix} x_2 & \ddot{q}_1 & x_4 & \ddot{q}_2 & x_6 & \ddot{q}_3 \end{bmatrix}^T
\end{aligned}
$$

B. Swing-Up

1) Energy Shaping: One approach to the swing-up of the pendulum is done by shaping the system's energy using passivity-based control and feedback linearization. We start by defining E_{pend}, which is the total kinetic and potential energy of both pendulum links. We also define E^* which is the potential energy of the pendulum at the up-up equilibrium point. Our goal is to drive E_{pend} to E^*. Currently, our system has the form:

$$
\begin{aligned}
\dot{\mathbf{x}} &= f(\mathbf{x}) + g(\mathbf{x})u \\
\mathbf{y} &= h(\mathbf{x})
\end{aligned}
$$

For reasons that be clear later, we employ input-output feedback linearization on the actuated dof to transform the system into the form:

$$
\begin{aligned}
\dot{\eta} &= f_0(\eta, \xi) \\
\dot{\xi} &= \mathbf{A_c}\xi + \mathbf{B_c}v \\
y &= \xi_1
\end{aligned}
$$

by using the following relationship for u:

$$
\begin{aligned}
u &= \alpha(\mathbf{x}) + \beta(\mathbf{x})v \\
\beta &= \frac{1}{L_g L_f^{r-1} h(\mathbf{x})} \\
\alpha &= -\frac{L_f^r h(\mathbf{x})}{L_g L_f^{r-1} h(\mathbf{x})}
\end{aligned}
\tag{8}
$$

Using Lie derivatives we find our system has relative degree $r = 2$ and we obtain:

$$
\dot{\xi} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ v \end{bmatrix}
$$

Note: we can now directly control the cart's acceleration with our new virtual input $v = \dot{x}_2 = \ddot{x}_c$.

In order to drive E_{pend} to E^*, we use the passivity-based control: feedback passivation. Given a nonlinear system, if the system is passive with a radially unbounded positive definite storage/Lyapunov function $V(x)$ and is zero-state observable then the origin x=0 can be globally stabilized by $u = -\phi(y)$. We try a Lyapunov function candidate of

$$
V(\mathbf{x}) = \frac{1}{2}(E_{pend} - E^*) \geq 0 \tag{9}
$$

which represents the difference between the pendulum's energy and the desired energy of the up-up equilibrium point. The idea here is to drive $V(x)$ to zero, which requires $\dot{V} \leq 0$. We get the following relationship:

$$
\begin{aligned}
\dot{V} &= (E_{pend} - E^*)\dot{E}_{pend} \\
&= (E_{pend} - E^*)\frac{\partial E_{pend}}{\partial \mathbf{x}}\dot{\mathbf{x}}
\end{aligned}
\tag{10}
$$

if we substitute $u = \alpha(\mathbf{x}) + \beta(\mathbf{x})v$ into the above relationship we get the following equation for $\dot{V}$:

$$
\begin{aligned}
\dot{V} &= (E_{pend} - E^*)\{v[(m_1 l_1 + m_2 L_1)x_4 \cos(x_3) \\
&\quad + m_2 l_2 x_6 \cos(x_5)] - (c_1 + c_2)x_4^2 - c_2 x_6^2 + 2c_2 x_4 x_6\} \\
&= (E_{pend} - E^*)\{vW - (c_1 + c_2)x_4^2 - c_2 x_6^2 + 2c_2 x_4 x_6\}
\end{aligned}
$$

It should be noted that if we had not performed feedback linearization, $\dot{E}_{pend}$ would be a very complex expression that would not simplify as nicely as it does here. By letting $v = -K(E_{pend} - E^*)W$ we get:

$$
\begin{aligned}
\dot{V} = {} & -K(E_{pend} - E^*)^2 W^2 \\
& - (c_1 + c_2)x_4^2 - c_2 x_6^2 + 2c_2 x_4 x_6
\end{aligned}
\tag{11}
$$

If we neglect damping, we can ensure $\dot{V} \leq 0$. In theory, we can now implement our control action u to drive the pendulum to the desired energy state, but in practice the cart overshoots and travels far from $x_1 = 0$. To combat this, we add a simple PD control term to v:

$$
v = -K(E_{pend} - E^*)W - k_1 x_1 - k_2 x_2 \tag{12}
$$

Now we can no longer mathematically guarantee passivity ($\dot{V} \leq 0$), but the results are validated through simulation.

While we now can drive E_{pend} to E^*, this does not guarantee the pendulum will balance at the desired position.

In fact, it will be swinging around wildly. We tune the gains K, k_1, k_2 so that the pendulum has enough energy to swing up, but at the lowest possible velocity, which lowers the performance requirements of the regulation controller.

2) Trajectory Optimization: Another approach to the swing-up phase is done by generating a pre-computed trajectory offline and tracking that trajectory online. To generate this trajectory, we solve the following optimization problem:

$$\begin{aligned} \mathbf{X}^*, \mathbf{U}^* = \arg\min_{\mathbf{x},\mathbf{u}} & \int_0^{t_f} \mathbf{u}^2 \, dt + \Phi(\mathbf{x}(t_f)) \\ \text{s.t.} \quad & \dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}) \\ & \mathbf{x}(0) = \mathbf{x}_0 \\ & \mathbf{x}(t_f) = \mathbf{x}_f \\ & u_{min} \leq \mathbf{u}(t) \leq u_{max} \end{aligned} \tag{13}$$

The optimization problem was reformulated and solved using a discrete-time NLP (trapezoidal direct collocation).

C. Regulation

To stabilize the up-up position, a linear model predictive controller (LMPC) is used. This controller solves the following optimization problem:

$$\begin{aligned} \mathbf{U}_k^* &= \arg\min_{\mathbf{U}_k} J(\hat{\mathbf{x}}_{k|k}, \mathbf{U}_k) \\ &= \arg\min_{\mathbf{U}_k} \sum_{i=0}^{N-1} \left[\hat{\mathbf{x}}_{k+i|k}^T \mathbf{Q} \hat{\mathbf{x}}_{k+i|k} + \hat{\mathbf{u}}_{k+i|k}^T \mathbf{R} \hat{\mathbf{u}}_{k+i|k} \right] \\ &+ \hat{\mathbf{x}}_{k+N|k}^T \mathbf{Q}_f \hat{\mathbf{x}}_{k+N|k} \\ \text{s.t.} \quad & \hat{\mathbf{x}}_{k+i+1|k} = \mathbf{A} \hat{\mathbf{x}}_{k+i|k} + \mathbf{B} \hat{\mathbf{u}}_{k+i|k} \\ & \hat{\mathbf{u}}_{k+i|k} \in \mathcal{U} \\ & \hat{\mathbf{x}}_{k+i|k} \in \mathcal{X} \end{aligned} \tag{14}$$

which can be reformulated using lifted dynamics:

$$\begin{aligned} \mathbf{U}_k^* = \arg\min_{\mathbf{U}_k} & \mathbf{U}_k^T (\mathcal{B}^T \tilde{\mathbf{Q}} \mathcal{B} + \tilde{\mathbf{R}}) \mathbf{U}_k + 2\mathbf{x}_k^T \mathcal{A}^T \tilde{\mathbf{Q}} \mathcal{B} \mathbf{U}_k \\ & + \mathbf{x}_k^T \mathcal{A}^T \tilde{\mathbf{Q}} \mathcal{A} \mathbf{x}_k \\ \text{s.t.} \quad & \hat{\mathbf{u}}_{k+i|k} \in \mathcal{U} \end{aligned} \tag{15}$$

IV. Results

A. Energy Shaping

A table of the tuning parameters is presented below, along with relevant figures.

TABLE II
ENERGY SHAPING PARAMETERS.

K	k_1	k_2	N	dt	$\mathbf{Q}$	$\mathbf{R}$	$\mathcal{U}$
5	5	3	30	0.01	diag[.1, .01, 15, 5, 15, .5]	1	[-50, 50]

In this setup, we switch from swing-up to regulation when $\theta_1, \theta_2 < 15\,\text{deg}$ and $\dot{\theta}_2 < 2$ rad/s, which occurs at $t = 19.1$s.

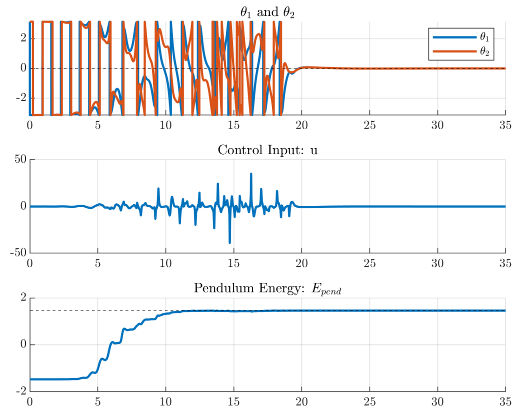

Fig. 2. Pendulum angles, control input, pendulum energy vs time

During the swing-up phase, the pendulum angles in Figure 2 are very chaotic. At this time, the pendulum links are swinging around wildly and their trajectory is uncontrolled. However, looking at subplot 3, we see our feedback passivation controller shaping the pendulum energy to the desired level. Despite the highly chaotic motion of the pendulums, we are able to precisely regulate the pendulums energy. Once switched at $t = 19.1$s to the regulation controller, we see virtual input v go to zero while the pendulum angles are stabilized at the up-up equilibrium point, which corresponds to the same desired E_{pend}.

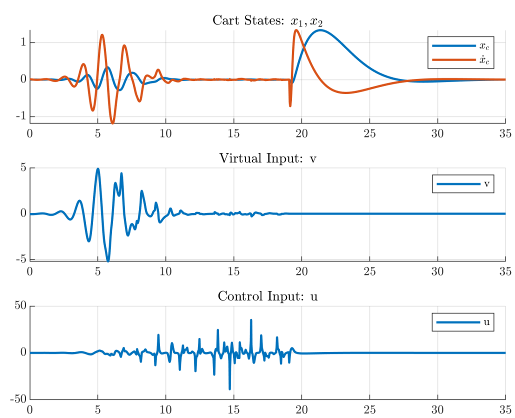

Fig. 3. Cart states, virtual input vs time

Figure 3 shows the cart states (pos/vel), virtual input v, and total input u. Recall from our input-output feedback linearization, we had $v = \ddot{x}_c$, that relationship is clearly seen in the first two subplots. Looking at the third subplot demonstrates the value of feedback linearization here, there is a simple relationship between the cart's dynamics and virtual input but clearly a very complex one between the cart and the total input.

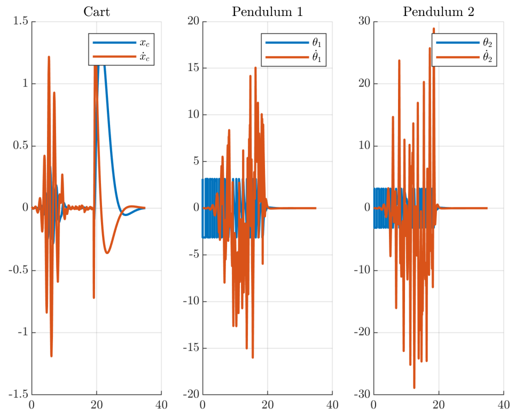

Fig. 4. System states vs time

Figure 4 shows the systems states, clearly chaotic during the swing-up phase, but stable during the regulation phase.

B. Trajectory Optimization

A table of the tuning parameters is presented below, along with relevant figures.

TABLE III
ENERGY SHAPING PARAMETERS.

N	dt	$\mathbf{Q}$	$\mathbf{R}$	$\mathcal{U}$
30	0.01	diag[.4, .01, 15, 5, 15, .5]	0.001	[-50, 50]

In this setup, we switch from swing-up to regulation when $\theta_1, \theta_2 < 2\,\text{deg}$, which occurs at $t = 4.26$s.

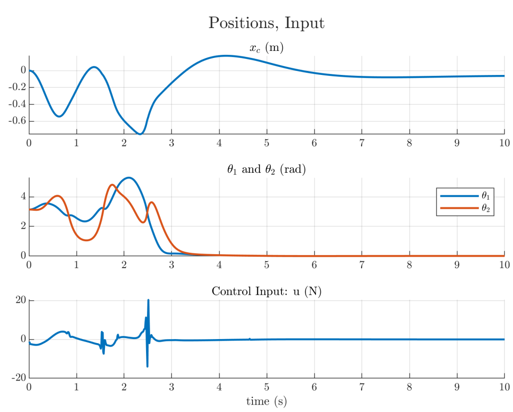

Fig. 5. Positions, input vs time

As seen in Figure 5, the system now takes a direct pre-planned path to the up-up equilibrium point, taking far less time to reach steady state than the energy shaping method.

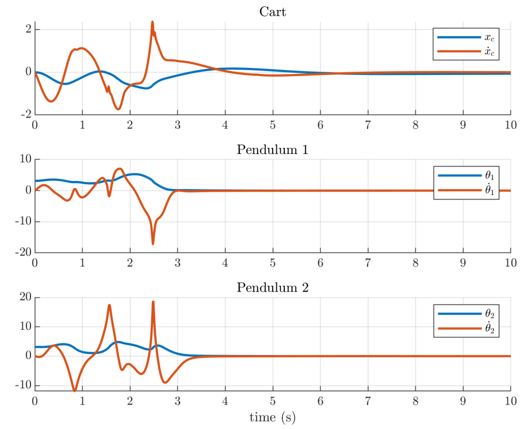

Fig. 6. System states vs time

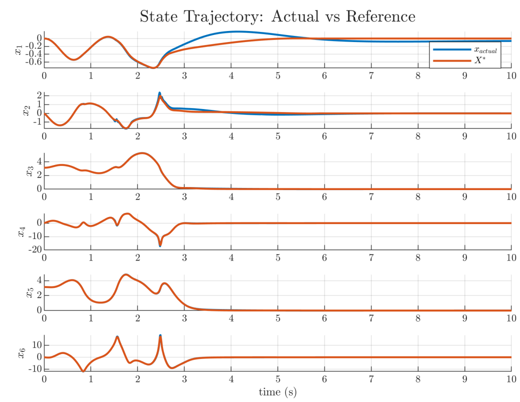

Fig. 7. States: Actual vs Reference

Figure 7 shows there is deviation between our true trajectory and the reference. The LMPC controller is still setup as a regulator and not a tracking LMPC. This is sometimes known as 'frozen-time MPC', and is likely the cause of these issues. At each time step, our MPC assumes that the reference trajectory stays constant at the X_k^* the optimizer is initialized with, and then tries to regulate the system to that value for the entire time horizon N. In reality, the reference is of course changing which results in the controller lagging behind reality.

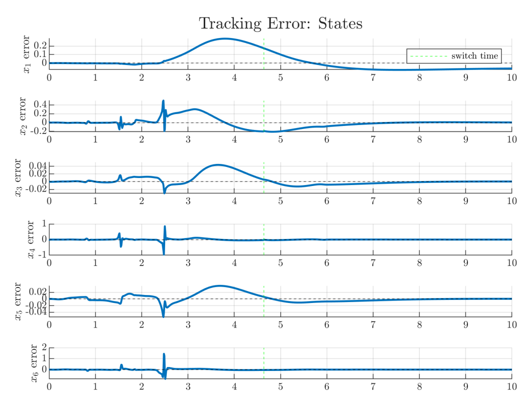

Fig. 8. State Error: Actual - Reference

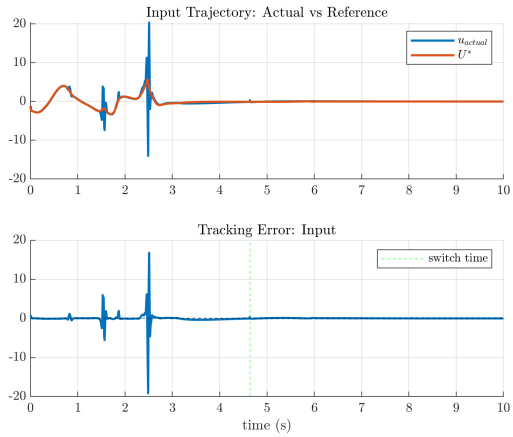

Fig. 9. Inputs: Actual vs Reference

Despite not using a tracking LMPC, and the deviation between reference and actual trajectory, it is still able to stabilize and perform well.

V. Conclusion

The project successfully analyzes two distinct swing-up strategies: Energy Shaping and Trajectory Optimization that are used to transition a double inverted pendulum from a stable "down-down" state to an unstable "up-up" equilibrium before handing control over to a LMPC.

The feedback passivation approach successfully drives the total pendulum energy to the desired up-up equilibrium level. However, as anticipated, this method is inherently chaotic, resulting in wild, uncontrolled pendulum trajectories during the swing-up phase. Furthermore, to prevent the cart from drifting excessively far from the origin, a PD control term had to be added to the virtual input. While this successfully kept the cart bounded, it sacrificed the mathematical guarantee of passivity. The energy shaping method required 19.1 seconds to maneuver the pendulums into the LMPC's Domain of Attraction ($\theta_1, \theta_2 < 15°$ and $\dot{\theta}_2 < 2$ rad/s).

The offline trajectory generated via trapezoidal direct collocation provided a highly controlled and deliberate swing-up motion. By tracking this pre-computed path, the system met the switching condition ($\theta_2 < 2°$) and stabilized in just 4.26 seconds.

For both swing-up methods, the transition to the LMPC regulation phase was successful. Once the switching conditions were met, the LMPC effectively stabilized the chaotic dynamics.

References

[1] W. Zhong and H. Rock, "Energy and passivity based control of the double inverted pendulum on a cart," in *Proceedings of the 2001 IEEE International Conference on Control Applications (CCA'01) (Cat. No.01CH37204)*, 2001, pp. 896–901.